OMIS 312: DATA GOVERNANCE – TERM PAPER

Data Governance Framework for Development: Empowering Ghana's Digital Future.

Sector Focus: Agriculture and Agri Businesses – A Case Study of Agromonti Company Ltd.

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EXECUTIVE SUMMARY

This case study examines Agromonti Company Ltd, a Ghanaian SME distributing fertilizers, seeds, and crop protection products to farmers nationwide. With a growing network of agronomists, logistics teams, and field agents, Agromonti plays a key role in Ghana's agricultural value chain by ensuring the timely delivery of essential inputs and technical support to both smallholder and commercial farmers. Despite its operational growth and data-rich environment, from GPS-tagged deliveries to field-level feedback, Agromonti currently operates without a formal data governance framework. Its data practices are largely manual and fragmented, exposing the company to inefficiencies, legal risks, and ethical concerns.

Internal research and a staff-wide questionnaire revealed critical gaps: unclear data ownership, lack of formal stewardship, weak enforcement of access controls, and no metadata tracking. While regular staff training and growing interest in AI/ML tools are promising signs, challenges reflect a broader issue common to SMEs in emerging economies, where informal practices often limit governance maturity more than technological constraints. The use of third-party data further raises concerns about traceability, data quality, and ethical handling.

To address these issues, the report proposes a Five-Pillar Data Governance Framework tailored to Agromonti's current structure: (1) clear definition of data roles and responsibilities, (2) a metadata and data quality registry, (3) tiered access control with encryption and logging, and (4) governance documentation integrated with business intelligence tools (5) compliance and consent Challenges. A 12-month roadmap outlines key steps, including staff workshops, data inventory development, automated quality checks, and breach simulations, all tracked through a performance dashboard.

Ethical considerations are also emphasized, particularly regarding customer consent and third-party data sharing. Recommendations include redesigning data forms to include consent checkboxes, publishing a privacy notice, and creating access channels for user data inquiries. To strengthen governance culture, initiatives such as a "Responsible Data Month" and the launch of a Governance Handbook are proposed. Long-term goals include achieving full metadata visibility, ensuring access control compliance, implementing real-time data quality dashboards, and fostering academic collaboration with institutions such as the University of Ghana Business School (UGBS).

In conclusion, data governance is essential to Agromonti's operational resilience, compliance, and digital transformation. With the right leadership, structure, and cultural support, Agromonti can become a model for responsible data management among SMEs in Ghana.

BACKGROUND AND CONTEXT

Data has become a fundamental asset across various fields of work. It is collected, analyzed, and leveraged to enhance financial optimization and strategic decision-making. As data becomes more integral to many aspects of business, it has become evident that rules and guidelines must be established to protect individual and personal data from use without prior consent. Recognizing this, the Government of Ghana enacted the "Data Protection Act, 2012," which seeks to protect the privacy of individual and personal data from misuse.

Organizations now have a legal and ethical responsibility to protect customer, individual, and employee data through the establishment of governance policy frameworks to safeguard all data related to the organization and its use. A notable example is the Facebook–Cambridge Analytica data scandal, where the personal data of millions of Facebook users was collected by the British consulting firm Cambridge Analytica for political advertising without informed consent (Facebook–Cambridge Analytica data scandal, 2025). This and similar events, such as Facebook's 2008 privacy breach scandal, underscore the dangers of weak or absent data protection frameworks. Consent.

In Ghana, many Small and Medium-Sized Enterprises (SMEs) have begun to integrate data usage into their everyday work processes, collecting more data on customers to drive sales and profits within their organizations. Yet most organizations lack and have failed to properly implement a well-structured policy to safeguard the data being handled by the organization, as demanded by the Data Governance Act (2012). Many organizations fall short of this due to factors such as a lack of knowledge, ignorance, and a blatant disregard for policy structure and its importance to the organization. Most SMEs are simply unaware of the existence of governance policies and the mandatory legal requirements for all organizations, along with the data they handle.

Data governance is defined as "the exercise of authority and control (planning, monitoring, and enforcement) over the management of data assets" (Ladley, 2012). It entails policies, procedures, roles, norms, and controls for the proper and ethical use of data to help an organization effectively accomplish its objectives (Eryurek et al., 2021). For SMEs, especially those competing in data-intensive markets like logistics and fintech, low data governance maturity can threaten both compliance and performance.

Data governance is not only about compliance; it also builds customer trust and loyalty. Research shows that customers are likely to remain loyal to organizations that are known to handle their data with the utmost care and protection. Global brands such as Starbucks and Amazon illustrate how well-managed data, governed with strong privacy protocols, regulatory compliance, and reliable security, enables them to consistently deliver personalized and smooth customer experiences (Baker, 2023).

Due to the strong link between data protection and customer loyalty, organizations are increasingly motivated to implement rigorous governance structures that ensure customer data is never accessed or used without prior consent. Data governance is a broad concept with no one-size-fits-all policy (Eryurek et al., 2021). While all organizations should implement it, the tools and processes must be tailored to suit that of a telecom company due to the nature of work

AGROMONTI DATA GOVERNANCE QUESTIONS AND ANSWERS WITH FRAMEWORK JUSTIFICATION.

ORGANIZATION & GOVERNANCE

Question	Response	Why the Framework is Necessary	Implementation Action
What is the organization hierarchy?	Board, MD, Consultants, Managing partners, etc.	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Who are the key decision-makers?	Board, MD, Technical team	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
How is performance evaluated?	3-month evaluation, sales rewards, mid-year, and end-of-year reviews	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
What is the current structure of your data governance team?	None	To eliminate ambiguity in data responsibilities and improve accountability.	Appoint a Data Governance Lead and formalize roles using a RACI matrix.
What are the specific roles of each governance team member?	None	To eliminate ambiguity in data responsibilities and improve accountability.	Appoint a Data Governance Lead and formalize roles using a RACI matrix.
What is your current company size (number of employees)?	30	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.

Do you use dashboards to monitor performance?	Yes	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Are automated reports generated from dashboards?	Yes	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.

DATA HANDLING & QUALITY

Question	Response	Why the Framework is Necessary	Implementation Action
What types of data do you collect?	Field staff, government/NGO data, Cocoa health, GPS, etc.	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
How is data collected?	By staff, government agencies, NGOs, and the sales agronomist team	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Where is the data stored?	On phones, manually in files, software, and backed up	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Who are data stewards?	Refer to the organogram: Director, Admin, Head of IT	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
What happens for data misuse?	Caution first, then dismissal	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.

Who owns/manages data in each department?	Administrator	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Are there clearly defined roles for data stewards/custodians?	Yes	To eliminate ambiguity in data responsibilities and improve accountability.	Appoint a Data Governance Lead and formalize roles using a RACI matrix.
How often is data reviewed or updated?	Weekly	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
What measures ensure ethical data use?	Audits, monitoring, and stewardship	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Are there staff guidelines/training for data handling?	Yes	Staff must understand governance to follow procedures correctly.	Hold quarterly governance workshops and ethical training.
How is data quality measured?	Observations, interviews, surveys, reports	Poor data quality affects business intelligence and decision-making.	Automate data quality checks and conduct weekly DQA reports.
What happens when incorrect data is found?	Leads to flawed decisions, ineffective operations	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Are there policies for data retention/disposal?	Yes – retention policy and disposal methods	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Is there a disaster recovery plan?	Yes	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.

METADATA & CATALOGUE

Question	Response	Why the Framework is Necessary	Implementation Action
Do you have a platform to manage metadata?	No	Absence of metadata leads to confusion and duplication.	Create a shared metadata registry using tools like Google Sheets.
Are data catalogues unified on one platform?	No	Absence of metadata leads to confusion and duplication.	Create a shared metadata registry using tools like Google Sheets.

REGULATORY COMPLIANCE & POLICIES

Question	Response	Why the Framework is Necessary	Implementation Action
How do you comply with Ghana's Data Protection Act?	Registered with DPC, implementing DPC principles	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Have you been audited before?	Yes	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Are formal governance policies in place?	Yes	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Who oversees policy implementation?	Director	To ensure legal compliance and reduce audit risks.	Register with the DPC and publish data governance policies.

Was your policy modeled after another organization?	No	To ensure legal compliance and reduce audit risks.	Register with the DPC and publish data governance policies.

DATA SECURITY & TECHNOLOGY

Question	Response	Why the Framework is Necessary	Implementation Action
What are your data protection policies?	NDAs, facial recognition, access control	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Who manages access?	Director and Head of IT	Lack of access control increases the risk of misuse and breaches.	Implement Role-Based Access Control (RBAC) and logging systems.
What happens during a cyber breach?	The IT team investigates, assesses, and resolves	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
What tools do you use?	OneDrive, Acronis, FACTURA ERP	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
What is your data retention period (cloud)?	2 years	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Do you manage access and permissions?	Yes	Lack of access control increases the risk of misuse and breaches.	Implement Role-Based Access Control (RBAC) and logging systems.

Are there alerts or systems to revoke access?	Yes	Lack of access control increases the risk of misuse and breaches.	Implement Role-Based Access Control (RBAC) and logging systems.
How do you share data securely?	Secure file transfer, clear agreements	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
What hardware/software do you use for data?	Sensors, scanners, databases, web tools	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Are these platforms cloud-based?	On-premises	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Are staff trained in data governance?	Yes	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
How often is training conducted?	Every 3 months	Staff must understand governance to follow procedures correctly.	Hold quarterly governance workshops and ethical training.

CHALLENGES & FUTURE GOALS

Question	Response	Why the Framework is Necessary	Implementation Action

What are your data governance goals?	Strengthen governance, improve quality, and foster a data-driven culture	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Suggestions for a data framework?	Use of AI/ML practices	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.
Current challenges?	Cost	Supports overall data governance maturity and ethical responsibility.	Incorporate into a phased implementation roadmap with clear KPIs.

GOVERNANCE AND STAKEHOLDER ANALYSIS.

Agromonti, like many SMEs, faces governance challenges driven by structure and necessity. Despite the ongoing need for enhanced, real-time data to support its logistics and agronomic services, the company currently lacks a comprehensive data governance framework. This results in various challenges, including inconsistent data quality, unclear access roles, ethical ambiguities, and compliance shortfalls. These issues pose significant risks for stakeholder engagement and hinder the ability to accurately measure performance.

OVERVIEW OF GOVERNANCE CHALLENGES AT AGROMONTI

The responses we've received point to several interconnected issues that are preventing the establishment of a robust data governance culture at Agromonti. These challenges can be attributed to five main themes: ambiguity of data roles and responsibilities, problems with data quality, a lack of policy documentation, uncertainty about compliance, and poor stakeholder alignment.

1. **Ambiguity of Data Roles and Responsibilities.**
One of the most significant gaps in governance is the absence of clearly defined roles and responsibilities for data management. Currently, the roles of data stewards seem to fall informally to the Administrator, Head of IT, and Director, yet there's no official organogram or charter to outline these governance duties. Access controls are managed through an unstructured hierarchy where admins generally have read-only access, while directors

and IT heads can both read and write. However, there isn't a formal access control policy in place to monitor data interactions, revoke access, or track changes. This lack of clarity puts the organization at risk for potential internal misuse, unauthorized alterations, and operational obstacles if one of these informal custodians were to leave unexpectedly.

2. **Data Quality Issues**.

Agromonti collects data from a variety of sources, including field agents, NGOs, and government bodies such as the Cocoa Health Extension Division of COCOBOD Ghana, as well as through mobile applications. The data encompasses field locations, GPS coordinates, sales transactions, stock status, and farmer information. While these sources provide a wealth of information, the quality of the data varies significantly. Issues such as data duplication, inaccuracies, outdated records, and conflicts between paper and digital formats present challenges for the company.

These problems stem from various factors: some result from human errors at the field level, others arise from the absence of validation procedures, and some are due to quality issues with partners, particularly institutions like IMANI Ghana. Although weekly audits are conducted, the absence of automated validation rules and error-checking systems allows many problems to persist until they affect reporting outcomes. As a result, Business Intelligence (BI) tools are rendered ineffective, limiting the company's ability to make timely decisions regarding logistics, sales routing, and fertilizer distribution.

3. **Lack of Policy Documentation and Centralized Framework.**

A major governance challenge at Agromonti is the absence of a clear and comprehensive data governance policy. There are currently no formal guidelines that define how data should be collected, processed, stored, or disposed of. Additionally, the company lacks a documented protocol for responding to data breaches. In a hypothetical scenario where a cyber breach occurs, the IT department's suggested response is to "contact the farmer, assess exposure, and resolve the issue." However, this approach lacks the necessary procedural format required by Ghana's Data Protection Act, 2012 (Act 843). Furthermore, the documentation issue persists because the company has not yet registered as a local data controller with the Data Protection Commission (DPC), failing to meet fundamental national legal obligations.

4. **Stakeholder Misalignment and Communication Gaps.**

Effective data governance hinges on the coordinated efforts of various stakeholders, both inside and outside the organization. However, insights gathered from field interviews highlight a troubling lack of alignment and communication among these key players. For example, sales agronomists often store important information on their mobile devices before it reaches IT. This practice results in delays, errors in data transmission, and uncertainty about who is responsible for what. Additionally, the absence of shared definitions for data and a lack of real-time information exchanges lead to inconsistencies in how different departments interpret and document critical metrics, such as the "number of farmers served" or the "product delivery date."

Moreover, third-party data processors, like government officials or NGO partners, aren't held to any data-sharing agreements or SLAs. This lack of contractual obligations leaves Agromonti unable to guarantee data accuracy, limit liability, or enforce appropriate security measures within the wider data ecosystem.

5. **Compliance and Consent Challenges**

Agromonti has laid some groundwork for data protection by including NDAs in employee contracts and storing analytical data using platforms like OneDrive and Acronis. However, field insights reveal that customers are not actively given the option to consent to data collection or to access, update, or withdraw their personal information.

The absence of privacy notices and formal retention schedules creates limited transparency in how data is managed across its lifecycle. Without clear communication, clients and partners remain uncertain about how their data is used, for how long, and under what conditions.

Moreover, treating all data the same, without accounting for sensitivity or legal requirements, could leave the company exposed to markets with stricter privacy regulations. Strengthening consent and retention practices will be critical to reducing risk and building long-term trust.

STAKEHOLDER ANALYSIS.

Agromonti has identified its key data stakeholders, recognizing the need to understand their interests and responsibilities to establish effective governance. A RACI matrix is used to clarify roles (Responsible, Accountable, Consulted, Informed).

a**. Board of Directors (Responsible)**

Sets strategy and ensures regulatory compliance, but lacks clear governance ties in performance metrics.

b. **Head of IT (Responsible and Consulted)**

Manages data systems and audits but lacks formal governance frameworks and resources for policy enforcement.

c. **Administrators and Operations Heads (Responsible and Informed)**

Control physical data access, but operates without defined roles or access controls.

d. **Sales Agronomists and Field Staff (Responsible and Well-Informed)**

Collect data but are unaware of data quality and privacy issues, often using insecure transmission methods.

e. **Partners - Government and NGOs (Consulted)**

Provide essential datasets but lack clear data-sharing agreements, risking low-quality inputs and breaches.

f. **Customers and Farmers (Informed)**

Own their data but have little control or visibility over its use, which erodes trust.

g. **Regulatory Agencies (Consulted and Oversight)**

The Data Protection Commission is not actively engaged, leaving Agromonti outside regulatory scrutiny, a gap that needs urgent attention.

Agromonti's data governance challenges are rooted not in negligence but in informal routines, inconsistent standards, and a lack of proper documentation. These issues affect all levels of the organization, from directors to field staff, and, if left unresolved, could lead to data misuse, regulatory breaches, and missed strategic opportunities. Nonetheless, the company has already begun adopting informal practices and has shown openness to technologies like AI and machine learning, signaling a readiness to strengthen its governance efforts.

This report therefore recommends a governance framework tailored to Agromonti's specific context, addressing stakeholder concerns while incorporating practical, best-in-class solutions suited to its operational environment.

PROPOSED FRAMEWORK

To build a saleable and sustainable Data Governance Framework for Agromonti Company Ltd, we must adopt a realistic approach that reflects the company's daily operations, regulatory landscape, and its current technological capabilities. The gaps in governance we have identified, such as ambiguous data roles, inconsistent data quality, lack of policy documentation and centralized framework, and informal access procedures, cannot be resolved by simply imposing a standard enterprise model. Instead, Agromonti needs a governance structure that is tailored to its size, workforce skills, and digital maturity.

This paper outlines a four-pillar Data Governance Framework, adapted from Ladley's DAMA-DMBOK2 model, incorporating insights gathered from Agromonti's field operations. This framework is designed not only to comply with Ghana's Data Protection Act, 2012 (Act 843) but also to facilitate Agromonti's ambitious vision for expansion throughout the West African sub-region.

Pillar 1: Definition of Data Roles and Responsibilities

The foundation of an effective governance framework lies in clearly defined roles. Currently, the IT Head, Director, and Administrator function as the informal custodians of Agromonti's information systems. However, the absence of formal responsibilities, authority levels, and reporting structures creates confusion over accountability.

To address this, the first pillar introduces a comprehensive governance charter that outlines roles and responsibilities. Key actions include:

a. Creation of a Data Governance Committee (DGC) of operations, logistics, and IT managers who would convene four times a year to address governance KPIs as well as policy modifications.

b. Appointment of a Data Governance Lead (DGL), preferably the IT Head, who will be responsible for overseeing data policies, access authorizations, and reporting of breaches.

c. Assignment of data stewards in key departments (Sales Agronomy, Finance, Administration) for regulating data quality as well as for communicating with the DGL.

d. Assignment of data owners, data custodians, data administrators, and data users with a clear definition of roles and responsibilities.

e. Use of a RACI matrix to define who is Responsible, Accountable, consulted, and informed for important governance activities.

According to Ladley (2012), one of the main reasons data governance initiatives fail is due to unclear ownership of the data. By documenting responsibilities, Agromonti can reduce reliance on individuals and improve accountability.

Pillar 2: Data Inventory and Metadata Management

Agromonti gathers information from field agents, NGOs, and partners using a combination of mobile apps, GPS, and paper reporting. However, the absence of a metadata registry and a shared inventory of data leads to confusion, data duplication, and misalignment in business intelligence.

This pillar focuses on creating a **Metadata Repository and Data Dictionary**, using accessible tools like Google Sheets. The registry will include:

a. Names and sources of datasets.

b. All data set owners and custodians.

c. Detailed descriptions and data types at the field level.

d. Rules for data validation and update frequency.

e. The status of the data (active, archived, deleted).

All stakeholders will be trained in how to use this registry to ensure consistent, accurate, and traceable data across departments.

Blokdyk (2020) emphasizes that overlooked metadata often causes "invisible failures" in business intelligence. A common data dictionary will significantly reduce such risks.

Pillar 3: Access Control, Security Protocols, and Data Breach Response

While Agromonti employs facial recognition for physical access and read/write controls for IT/management, neither formally written access rights nor response procedures for a breach are in place. This pillar introduces **scalable security measures across three areas**:

a. Role-Based Access Control (RBAC):

i. Assign roles and provide data access levels (e.g., read only, read/write, admin).

ii. For Microsoft OneDrive and Acronis platforms, apply tiered access.

iii. Maintain access logs and require two-factor authentication for sensitive material.

b. Breach Response Plan:

i. Establish procedures for detecting, reporting, and resolving data breaches.

ii. Include IT, Communications, and Legal Compliance roles in responding to an incident.

iii. Create notice templates for impacted individuals as well as regulators (e.g., the DPC).

c. Data Classification Policy:

i. Categorize datasets by sensitivity (e.g., personal, transactional, partner-sensitive)

ii. Relate classification levels with appropriate retention, encryption, and access procedures.

The Facebook–Cambridge Analytica case shows what can happen because of insufficient access controls and a lack of a breach notice process (Isaak & Hanna, 2018). Agromonti can sidestep a similar problem with proportional measures.

Pillar 4: Aligning Business Intelligence (BI) and Governance through Automation.

Agromonti plans to use BI dashboards to drive decisions in logistics, fieldwork, and sales. However, without aligning governance protocols, dashboards can mislead or confuse users. To address this, we recommend the following steps:

a. Link BI measures to certified data sources documented in a metadata registry.

b. Incorporate data quality indicators in the dashboards, such as "Last updated: 5 days ago," and "Data accuracy score: 92%."

c. Automate data quality validation processes using tools like Google AppScript or Python scripts, focusing on identifying anomalies, duplicates, and missing values on a weekly basis.

Agromonti's interest in AI and machine learning adds a progressive dimension to this pillar. After establishing robust governance practices, Agromonti can investigate:

a. Utilizing predictive analytics for inventory planning.

b. Implementing AI-driven anomaly detection for quality control.

c. Developing chatbot interfaces for farmers to edit data or request services.

Drawing inspiration from Netflix's strategy of embedding governance in its algorithmic
Processes demonstrate that advanced AI systems can be rooted in ethical and transparent governance principles

(Eryurek et al., 2021). Agromonti can adopt a simplified version of this approach, utilizing tools that are accessible to subject matter experts.

Pillar 5: Ethical Data Use and Consent Management

Agromonti collects personally identifiable information (location of growers, telephone numbers, payment history), but makes no clear choices for data subjects of consent, access, or correction. This pillar fills this ethical and legal gap with:

1. Implementation of Consent Forms and Notices of Privacy at all collection points (physical and digital), in plain language and in a language as needed.

2. Digitized consent logging with data-stamped digital applications (e.g., mobile data applications, Google Forms).

3. Development of a Customer Data Rights Policy outlining how users can:

a. Request access to their data

b. Correcting inaccuracies

c. Withdraw consent

d. Report misusing or filing complaints

The fifth pillar entails education of all employees on the significance of minimizing data, where only necessary data should be acquired, stored, as well as utilized.

Ethical governance starts with the data entry point as noted by Boateng (2022), and should be maintained all the way to retention, usage, and deletion. SMEs should look beyond the convenience of operations to ensure adherence to law as well as morality.

FRAMEWORK VISUALIZATION.

Pillar	Objective	Tools	Lead Role

Role Definition	Assign and document stewardship	RACI matrix, charter	DGL (IT Head)
Metadata Registry	Define, standardize, and track data assets	Google Sheets, Airtable	Data Stewards
Access & Security	Secure access, log activity, and handle breaches	RBAC, 2FA, Incident Protocol	IT + Admin
BI & Automation	Align data quality with decision-making	Dashboards, scripts, ML	Analytics Team
Consent & Ethics	Ensure ethical and legal data	Consent forms, Rights policy	DGL

In summary, this five-pillar framework provides a structured yet scalable solution for Agromonti to transition from informal data practices to institutionalized governance. It emphasizes cost-effective tools, staff empowerment, and legal alignment, making it both implementable and sustainable in an SME environment.

IMPLEMENTATION ROADMAP

The proposed data governance framework will be rolled out in four progressive phases over a 12-month period. Each phase is designed to build upon the last, moving from foundational setup to long-term cultural adoption.

Phase 1 (Months 1–3): Setting Foundation and Policy Making

Primary Objectives:	**Activitity**	**Output:**

Establish governance, leadership, and structure.	Select a Data Governance Lead (DGL)	Signed governance charter
Conduct baseline audits and risk assessment.	Establish Data Governance Committee (DGC)	Specified roles and responsibilities
Develop core policies and definitions of roles	Conduct a Data Governance Maturity Assessment	Baseline maturity report
	Prepare First Policy Papers	Commenced DPC activity (registration as a controller)
	Staff Orientation Workshop Conduct	.

Phase 2 (Months 4–6): Operational Setup and Quick Wins

Principal Objectives	**Activities**	**Output**
Implement the first governance controls.	Use Role-Based Access Control (RBAC)	Operational metadata repository
Develop an asset and metadata registry.	Create Metadata and Data Asset Inventory	Functional RBAC across platforms
Commence minimal quality tests and consent procurement	Begin Infrastructure for Consent Collection	Privacy statements and consent histories in use
	Report weekly Data Quality Assessments (DQAs)	First wave of DQA reports

	Train Frontline Staff on Quality and Ethics	

Phase 3 (Months 7–9): Process Automation and Compliance Integration

Primary Objectives	**Activities**	**Output**
Simplify recurring governance tasks.	Automate Data Quality Monitoring	Automated governance alerts
Enable dashboards for governance KPI tracking.	Start Governance Dashboard	On-screen KPI dashboard
Authenticate responses to incidents.	Conduct Incident Response Simulation	Signed DSAs with key partner
	Begin External Partner Engagement	Intentional software vulnerabilities exploited.
	Submission of Annual Compliance Brief to DPC	

Phase 4 (Months 10–12): Culture Building and Expansion

Primary Objectives	**Activities**	**Output**
Institutionalize governance culture	Launch a “Responsible Data Month” Campaign	Annual governance awards

Measure progress and scale up tools.	Create Governance Handbook	Completed governance handbook
Future-proof for long-term viability	Final Review and Scorecard Evaluation	Revised maturity report
	Begin Planning for Integrating AI	AI readiness document
	Develop Governance Roadmap 2.0	

OPERATIONAL SUPPORT MODEL FOR IMPLEMENTATION.

Support Component	**Frequency**	**Responsibility**
DGC Meetings	Monthly	Data Governance Lead
DQA Reports	Weekly	Data Stewards
Training Sessions	Quarterly	Admin/HR
Policy Reviews	Biannual	DGC
External Audits	Annual	IT + Director

TOOLS AND TECHNOLOGY

The Agromonti blueprint is designed with low-cost, accessible software:

Function	**Tools/Platforms**
Storage & Backup	Microsoft OneDrive, Acronis
Documentation	Google Sheets, Airtable
Consent Collection	Google Forms, Mobile Data Apps
Automation	Google AppScript, Python Scripts
Dashboards	Google Data Studio, Power BI
Communication	WhatsApp Groups, Email, Google Docs

This approach to implementation isn't just practical; it's truly transformative. It addresses Agromonti's actual readiness and the realities on the ground, leveraging strengths, addressing weaknesses, and ensuring that the business aligns with both ethical and legal data practices. If fully embraced, this strategy has the potential to set. Agromonti apart as a regional standard for SME-level data governance and leadership.

RISK AND ETHICAL CONSIDERATION

Agromonti must assess both its current data practices and future ambitions through the lens of risk and ethics. Although positive steps have been taken, such as using cloud storage, facial recognition, and staff training, significant gaps remain in governance, legal compliance, operations, and ethical data use.

If ignored, these issues could impact efficiency, trust, and growth. This section outlines key risks and offers solutions based on lessons from global cases like the Facebook–Cambridge Analytica scandal and companies like Netflix.

1. **Strategic Governance Risks**

Informal governance structure is one of the biggest Agromonti risks since it creates ambiguity in leadership, decision-making, and reporting lines. Key individuals such as the Administrator, Director, and Head of IT currently serve custodial roles, but their responsibilities aren't documented in any governance policy. This exposes the business to:
a. Failure to manage roles properly, particularly for positions with significant responsibilities
b. Paralysis of operations should a key steward vacate or be indisposed
c. Ineffective enforcement due to the informal nature of authority

Until governance gains prominence at the strategic level, it will be difficult to justify investments in policies, tools, or training.

Mitigation Strategy:
a. Framing governance performance indicators (e.g., consent coverage, access violations) into quarterly review meetings
b. Periodic reporting to Managing Director and Board by the Data Governance Committee (DGC)

2 **Operational and Data Quality Risks**

Agromonti's data operations face various risks that affect decision-making and damage reliability:
a. Mixed datatypes from field workers, often sent over mobile devices without standardization
b. Manual input leading to duplicate entries and missing values
c. Incorrect details from third-party partners like IMANI Ghana, with no agreed quality control
d. These issues impair BI performance and affect customer service, inventory, and routing decisions

Mitigation Strategy:
a. Introduction of automated data quality scripts to flag anomalies
b. Weekly Data Quality Assessments (DQAs) by trained stewards
c. Signing of Data Sharing Agreements (DSAs) with all external partners to enforce data standards

3**. Legal and Regulatory Compliance Risks**

As a Ghanaian SME handling personal data of farmers, employees, and partners, Agromonti is required to comply with the Data Protection Act, 2012 (Act 843). Failure to do so could result in sanctions, reputational damage, or legal action. Current gaps in compliance include:
a. Not being registered with the Data Protection Commission (DPC)
b. No formal consent process for farmers and buyers

c. Absence of a breach response plan
d. No system for customers to access or correct their personal data

Mitigation Strategy:
a. Register promptly with the DPC and submit annual compliance reports
b. Implement a centralized consent management system
c. Create a standardized breach response plan

4. Ethical Risks: Consent, Clarity of Information

Beyond legal compliance, Agromonti must address the ethical aspects of how data is collected and used.
a. Farmers and field agents aren't fully informed about how their data is handled
b. There are no written privacy notices or clear opt-out options
c. Internal staff lack clarity on data-sharing boundaries, which raises risks of overcollection or misuse, especially in areas with low digital literacy

Mitigation Strategy:
a. Provide simple, multilingual privacy notices at all data collection points
b. Train staff on ethical data use and the principle of data minimization
c. Add data ethics to quarterly staff refresher training

5. Emergent Threats of AI and Future Challenges of Governance

As Agromonti explores the use of AI/ML technologies for forecasting, route optimization and quality verification, it must also prepare for the ethical and governance risks these systems can introduce. If left unchecked, AI may cause unintended harm or regulatory backlash. Potential challenges include:

a. Amplifying decision-making biases due to biased or incomplete training data
b. Obscuring transparency if algorithmic reasoning is hidden from users or stakeholders
c. Violating privacy rights, especially through profiling without user consent

Mitigation Strategy:
a. Apply AI Ethics Checklists to ensure fairness, transparency, and accountability in all models
b. Conduct lightweight Ethical Impact Reviews (EIRs) before deploying AI-powered features
c. Require human oversight for all sensitive decisions made by AI systems

For Agromonti Company Ltd, risks are not only obstacles but because of the reasons where governance needs to change courses. The business stands at a crossroads: take the path of semi-form all controls and accommodate

increasingly larger exposure or take an active hand in developing a culture of accountability, compliance, and ethical leadership.

Preparing for and responding to these risks for the future, Agromonti can maintain customer trust, comply with regulatory expectations, and innovate safely in its applications of technology. We will outline in the next section some fundamental strategies with a vision for Agromonti as a data-responsible SME in Ghana, as well as internationally.

CONCLUSION AND STRATEGIC RECOMMENDATIONS

Agromonti Company Ltd is at a pivotal point in its journey as a data-driven agro-logistics SME in Ghana. Although the company generates rich data from field operations, partnerships, and logistics activities, its current governance practices are largely informal, fragmented, and not aligned with regulatory or strategic standards. This leaves Agromonti exposed to inefficiencies, poor data quality, legal vulnerabilities, and ethical risks, especially as it aspires to scale across West Africa.

This report presented a structured, scalable Data Governance Framework rooted in four pillars, role definition, metadata management, access and security, and alignment with BI and automation. A 12-month phased roadmap, implementation support model, and risk mitigation strategy were also outlined. The goal is to move Agromonti toward full data governance maturity in a practical, cost-conscious, and regulation-aligned manner.

To begin, Agromonti should formally appoint a Data Governance Lead (DGL) with the authority defined in a comprehensive Governance Charter. This lead should be supported by a cross-functional Data Governance Committee (DGC) made up of representatives from sales, logistics, IT, and administration. The committee should embed governance into day-to-day operations through quarterly KPI reviews focused on metrics like consent coverage, access violations, and data quality scores.

Compliance with Ghana's Data Protection Act, 2012 (Act 843) must be prioritized. Agromonti should register as a data controller with the Data Protection Commission (DPC) and designate a compliance specialist to oversee regulatory engagement, breach responses, and annual audits. This ensures both legal alignment and transparency in the company's data operations.

To support traceability and standardization, the company should build a metadata repository and data dictionary using affordable tools like Google Sheets or Airtable. These registries should detail dataset names, custodians, update frequencies, and usage, and they should be linked directly to BI dashboards for transparency in decision-making.

In redesigning data collection, Agromonti must ensure explicit consent is collected, via digital checkboxes or physical forms, with clear, multilingual privacy notices. A "Know Your Rights" inquiry system (webpage or hotline) should be introduced to empower data subjects to access, rectify, or erase their data.

Role-Based Access Controls (RBAC) must be enforced across platforms like Microsoft OneDrive, with log tracking and two-factor authentication enabled for sensitive datasets. Weekly audits should monitor for unauthorized access, supporting data protection and accountability.

To enhance operational efficiency, automated scripts using Google AppScript or Python should perform regular data quality checks. Weekly DQA reports should be disseminated to staff, creating a culture of openness and continuous improvement. Agromonti must also formalize relationships with external partners through Data Sharing Agreements (DSAs). These contracts should outline responsibilities for data accuracy, retention periods, and breach reporting to minimize third-party risks and support lawful processing.

As the business explores AI and machine learning applications, Ethical Impact Reviews (EIRs) must be conducted before deploying any models. Agromonti should ensure fairness, model transparency, and human oversight to mitigate bias and build user trust.

Finally, fostering a governance-driven culture is key. The launch of a "Responsible Data Month" and a Governance Handbook across all departments will institutionalize best practices and raise internal awareness. In the long term, Agromonti should aim to publish its governance progress and collaborate with academic institutions like the University of Ghana Business School to share lessons and scale impact.

In conclusion, these strategic recommendations form a practical, actionable blueprint tailored to Agromonti's unique context as an SME. If adopted with discipline and purpose, they will transform Agromonti from a data-rich but vulnerable entity into a beacon of ethical, compliant, and future-ready data governance leadership in Ghana and across Africa.

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APPENDIX

A: Agromonti Field Data Summary.

Primary Data Sources:

1. Responses to structured questionnaires from the Head of IT, Director, Administrator, and Sales Agronomists

2. Follow-up interview transcripts

3. Internal documents on employee evaluation, access protocols, and storage policies

Data Collection Methods:

1. Structured questionnaires

2. Semi-structured interviews

3. Submission of digital forms (Google Forms and PDFs)

4. Observations from site visits and mobile field surveys

Technologies Used at Agromonti:

- **Microsoft OneDrive** – Primary data storage

- **Acronis** – Cloud backup solution

- **Facial Recognition** – Access control system

- **Mobile GPS Devices** – Location-based tracking for logistics

- **Google Sheets and Paper Logs** – Used for manual and digital data entry

- **Manual Signing of Employee NDAs** – For internal confidentiality and accountability

Appendix B: Metadata Registry Template (Proposed)

Dataset Name	Department	Owner	Format	Frequency	Description	Last Updated
Farmer Profiles	Agronomy	Sales Team	CSV	Weekly	Farmer name, location, inputs	2025-07-15
Sales Reports	Finance	Accountant	Excel	Monthly	Branch sales, receipts, product	2025-07-01
NGO Field Reports	Programs	Program Lead	PDF	Quarterly	Activity summaries by partner	2025-06-30

Appendix C: Recommended Governance Dashboard KPIs

- Percentage of Metadata Completion – Tracks documentation of data assets
- Percentage of Consent Forms Collected This Month – Measures data subject compliance
- Average Data Quality Score (Weekly) – Based on validation scripts and steward reviews
- Number of Access Incidents (Last 30 Days) – Indicates breach or misuse risk
- Percentage of Staff Trained per Course Offered – Monitors data literacy and governance education

Appendix D: Ethical Impact Review Questions for Readiness for AI Use

1. On what dataset its model trains. If balanced.
2. Is the model biased in its output?
3. Are decisions understandable for affected users?
4. Is there a human override capability?
5. Is customer consent provided prior to profiling?

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